
SUMMER CAMP

小練習



小練習第一題

PROMPT ENGINEERING

任務介紹 - PROMPT ENGINEERING

問：教我做蛋炒飯。



答：

準備白飯、雞蛋、蔥花……

熱鍋下油……

翻炒均勻即可。

問：你是一位主廚，請用最簡單的方式教我做蛋炒飯，只能使用 5 種食材，每一步不超過一句話。



答：

食材：白飯，雞蛋，蔥，油，鹽

步驟：

先把雞蛋打散。

熱鍋放油，炒熟雞蛋。

加入白飯炒散。

加入鹽和蔥花拌勻。

盛盤即可。

我們的小練習 - 請各位當助教

我們準備了一題在上課作業中給同學們回答的簡答題，會提供20個同學的答案，有些答得好有些在瞎掰，請各位：

- 用指揮AI的方式改作業
- 運行我們提供的程式，實現自動打分



問題介紹

我們當時的作業是用LSTM (RNN) 模型做加減乘除題，題目是：

- Why do we need gradient clipping during training? 這題的滿分是5分

Gradient clipping是一個防止模型訓練的時候發生梯度爆炸的方法。

- 為了回答這個問題，我們需要知道基本思路：

1. 什麼是Gradient clipping?

- Gradient clipping 是一種在梯度太大的時候的時候將其縮小的機制。

2. Training的時候發生了什麼?

- 在 LSTM 訓練時，由於長序列的梯度反覆相乘，可能造成 gradient explosion (梯度爆炸)，使梯度變得非常大，導致模型參數更新失控。

3. Gradient clipping如何解決training時發生的現象?

- Gradient clipping 會在梯度過大時把它縮小到合理範圍，讓參數更新保持穩定，避免訓練發散。

給個例子

- Why do we need gradient clipping during training?
- Gradient clipping 有點像是在做 Linear Regression時對 Loss function 做的 Regularization，對很大的權重做懲罰，避免 Overfitting。Gradient clipping 則是對每次更新的梯度進行限制，避免 RNN 這種序列的模型在訓練的過程中，出現某個大於1的梯度時就會在後續的多次連乘後變得超級大，也就是梯度爆炸，讓原本穩定下降的 loss 突然就變得超大甚至變成 NaN，導致模型訓練失敗。

給個例子

- Why do we need gradient clipping during training?
- 答：Gradient clipping 有點像是在做 Linear Regression時對 Loss function 做的 Regularization，對很大的權重做懲罰，避免 Overfitting。Gradient clipping 則是對每次更新的梯度進行限制，避免 RNN 這種序列的模型在訓練的過程中，出現某個大於1的梯度時就會在後續的多次連乘後變得超級大，也就是梯度爆炸，讓原本穩定下降的 loss 突然就變得超大甚至變成 NaN，導致模型訓練失敗。

這個答案是滿分5分

再給個例子

- Why do we need gradient clipping during training?
- 答: Clipping is a classic method of preventing gradient explosions.

再給個例子

- Why do we need gradient clipping during training?
- 答：Clipping is a classic method of preventing gradient explosions.

這題的打分是2分

原因：未詳細說明梯度爆炸的成因以及梯度裁減是如何處理梯度爆炸的

評價指標 - MSE

- 我們會以當時的助教們手動批改的作業為標準答案，計算各位的預測結果和答案直接的MSE（Mean Square Error），這個數值越低越好。

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

標準答案

預測結果

小練習如何進行

這次小練習我們用Kaggle的比賽來進行

- 各位可以單人進行，也可以組隊（最多5人）
- 最後我們會留10分鐘，請前三名上來介紹一下自己的做法。

給一些參考思路

1. 指定角色

- 比如：你是一位老師...

2. 把複雜的問題拆解成一些小問題

3. In context learning (Few shot)

- 在寫Prompt的時候提供一些範例。
- 我們有提供一個few_shot.csv，裡面有五筆資料可以作為範例。

4. 用中文/英文寫prompt

KAGGLE

- Kaggle: <https://www.kaggle.com/>

搜尋: [T1] NTHU Summer Camp 0723

或者直接進入鏈接: [LINK](#)

The screenshot shows the Kaggle homepage for user FengZQ. On the left is a sidebar with navigation links: Home, Competitions, Benchmarks, Game Arena, Data Hub, More, Your Work, and Viewed. Below these are several project entries, including '[T1] NTHU Summer C...', '[T2] NTHU Summer ...', and '0723 NTHU Summer ...'. The main content area features a search bar at the top, which is highlighted by a red arrow. Below the search bar is a welcome message 'Welcome, FengZQ!' and a progress indicator showing '5 days a new record!' and '0% to Expert'. A row of statistics follows: Datasets (8 total created), Notebooks (26 total created), Competitions (42 total joined), Discussions (0 total posted), and Courses (0 total completed). At the bottom, there is a section titled 'How to start: Choose a focus for today' with three cards representing different focus areas: Competitions, Courses, and a third card with a question mark icon.

加入比賽

進入這裡可以看到我們準備好的代碼

 FENGZQ · COMMUNITY PREDICTION COMPETITION · 2 DAYS TO GO

[T1] NTHU Summer Camp 0723

This competition provides a question and 20 corresponding student answers. Participants are required to grade the answers.

[Overview](#) [Data](#) [Code](#) [Models](#) [Discussion](#) [Leaderboard](#) [Rules](#)

Overview

歡迎來到清大資工營！

在本次的小練習中，你將透過挑戰真實世界的自然語言處理任務：「自動化簡答題評分（ASAG）」，來探索大型語言模型（LLM）的實際應用。你的目標是設計、測試並優化提示詞（Prompt），指示 LLM 成為一個公平且準確的自動評分員。

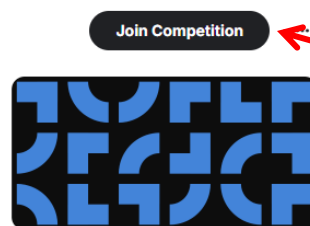
Welcome to the NTHU CS Summer Camp!

In this exercise, you will explore the practical applications of Large Language Models (LLMs) by tackling a real-world Natural Language Processing task: Automated Short Answer Grading (ASAG). Your objective is to design, test, and refine prompts that instruct an LLM to act as a fair and accurate automated grader.

Start
6 hours ago

Close
2 days to go





點擊此處加入比賽，用 google 帳號登入

Competition Host
FengZQ 

Prizes & Awards
Kudos
Does not award Points or Medals

Participation
0 Entrants
0 Participants
0 Teams
0 Submissions

Tags
Mean Squared Error

Table of Contents 

組隊

Overview Data Code Models Discussion Leaderboard Rules **Team** Submissions

Let others know you're looking for teammates

[See tips about forming teams](#)

☐ Display your status with  on the leaderboard

Team Members

Your team can have a maximum of 10 members.



Zhi-Quan_Feng (You)
Team Leader

Send Invitation

You can invite 9 more people to join your team. They will need to have joined the competition.

 Search for the team or user name

Received Invitations

You haven't received invitations from other teams.

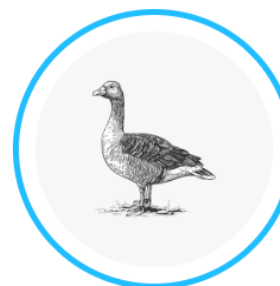
Save Changes

從這裡進入可以
和別人組隊

搜索他人的ID就可以拉
人入隊（每隊最多5人）

點擊頭像進入個人頁
面，就可以看到ID

ID在這裡



zhiquanfeng

Zhi-Quan_Feng

 Joined 16 minutes ago · last seen in the past day

打開NOTEBOOK

 FENGZQ · COMMUNITY PREDICTION COMPETITION · 2 DAYS TO GO

Submit Prediction

...

[T1] NTHU Summer Camp 0723

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Overview Data **Code** Models Discussion Leaderboard Rules Team Submissions Settings

Notebooks

+ New Notebook

Filters

All Your Work Shared With You BookmarksHotness

Pinned notebooks

 NTHU Summer Camp Sample Code 1

Updated 4h ago

0 comments · [T1] NTHU Summer Camp 0723

2

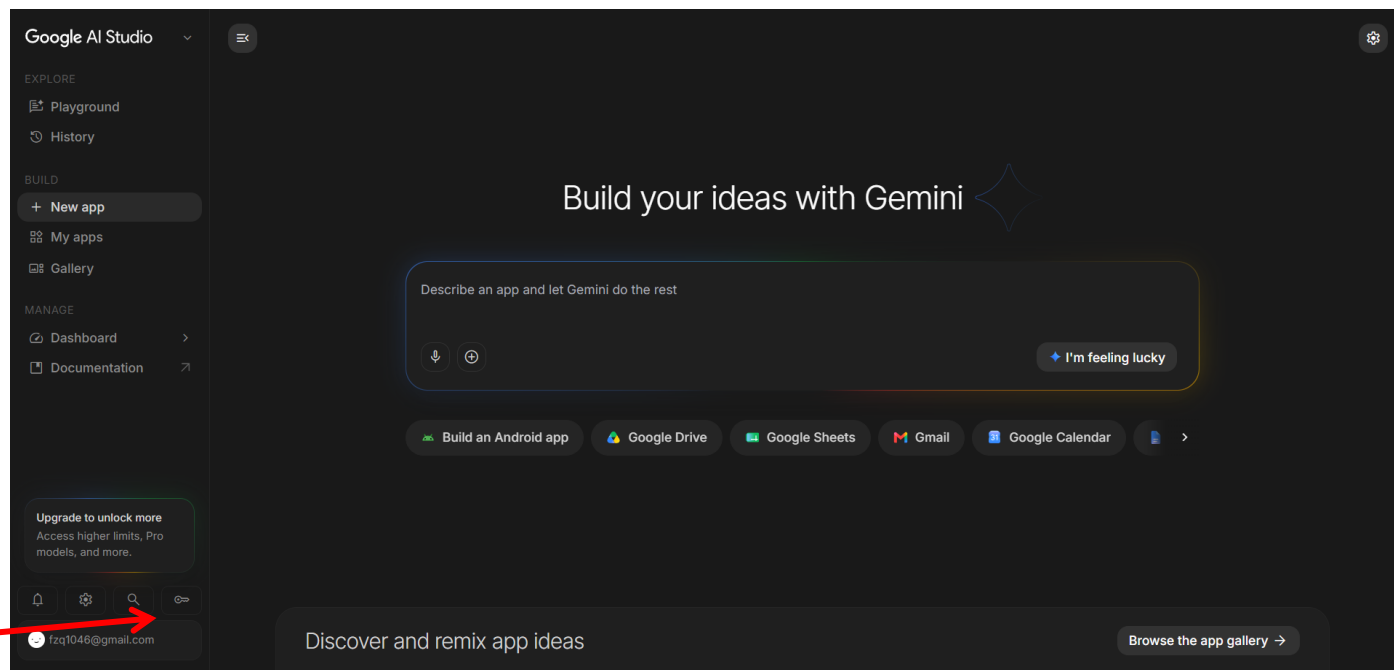
...

進入這個notebook，就可以直接運行代碼

GOOGLE AI STUDIO

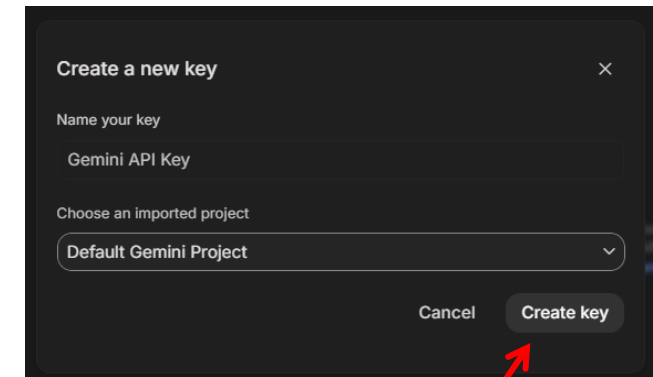
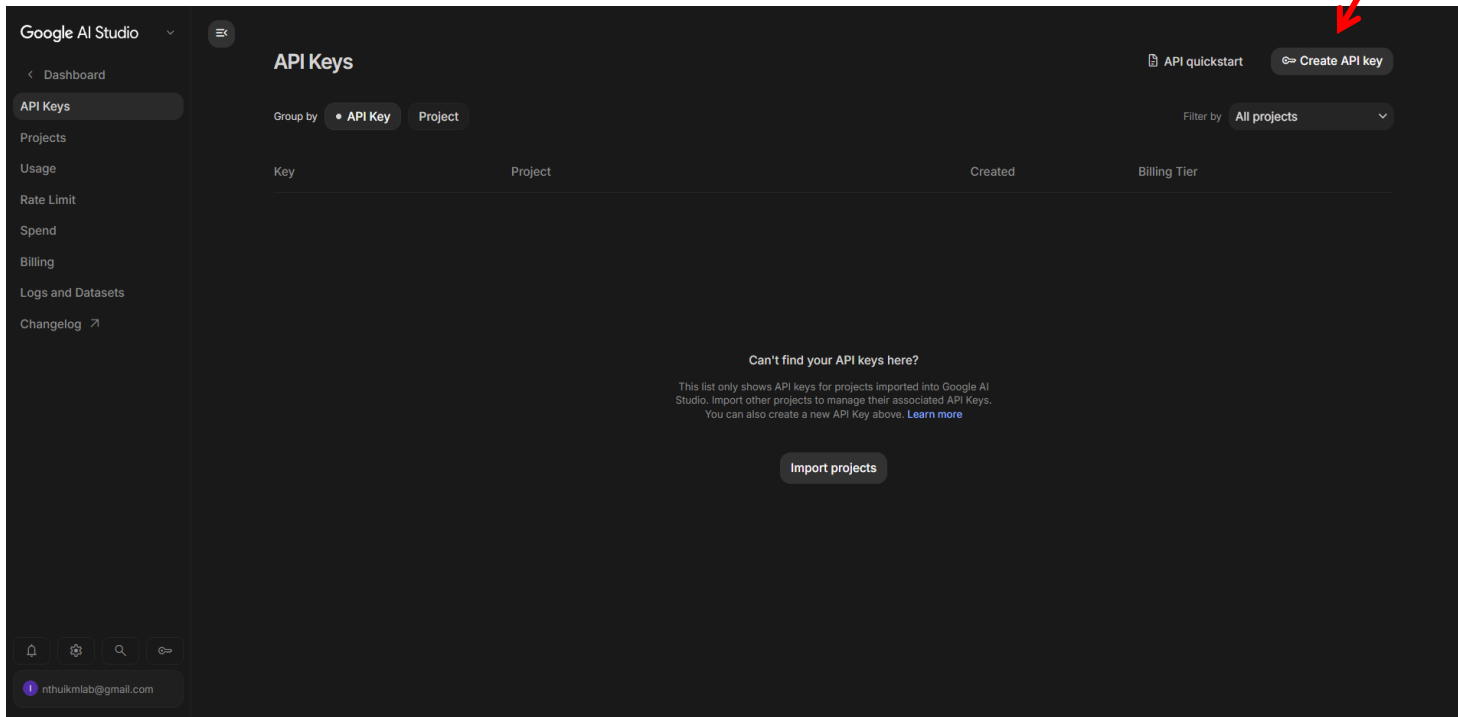
Google 直接搜索 Google AI Studio, 或者直接進入: [LINK](#)

點擊此處進入，創建
Access Token



ACCESS TOKEN

點這裡

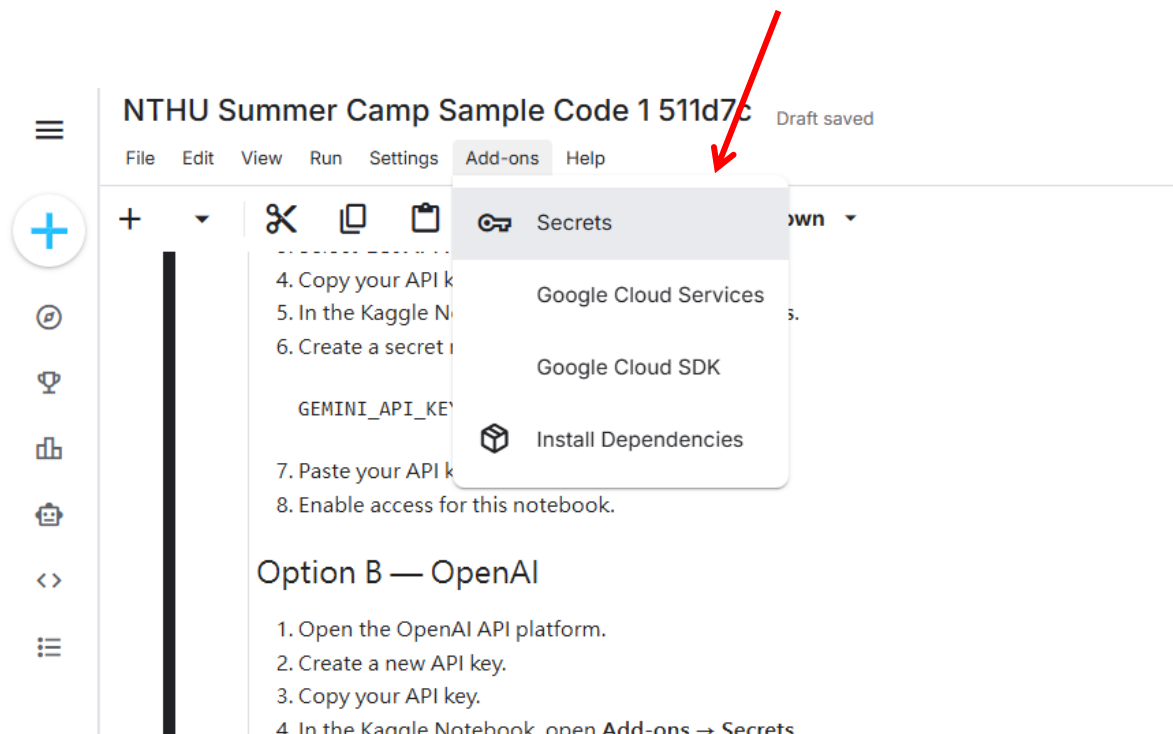


點這裡

然後複製建立出來的 Access Token

ACCESS TOKEN

從這裡導入Access Token



Secrets



No secrets added
Add secrets like API keys

Add secret

點這裡

命名為GEMINI_API_KEY
然後把Access Token貼上去

Secrets

New secret

LABEL

Enter label

VALUE

Enter value

Save

Cancel

PROMPT ENGINEERING

```
# =====  
# THIS IS THE MAIN PART THAT PARTICIPANTS SHOULD EDIT  
# =====  
  
PROMPT_TEMPLATE = """  
You are a teacher grading a student answer.  
  
Read the question and the student answer. Then give a score from 0 to 5.  
  
Rules:  
1. Use only the information provided.  
2. A score of 0 means the answer is completely wrong or does not answer the question.  
3. A score of 5 means the answer is correct, complete, and well explained.  
4. Lower the score when the answer is missing important information, evidence, or clear reasoning.  
5. Return exactly one JSON object in this format:  
   {{\"score\": 3}}  
6. Do not use Markdown.  
7. Do not write anything outside the JSON object.  
  
QUESTION:  
{question}  
  
STUDENT ANSWER:  
{student_answer}  
"""
```

找到代碼中間的這段，這裡的文字部分就是給模型的prompt，裡面的內容可以任意修改



執行程式

NTHU Summer Camp Sample Code 1 511d7c Draft saved

File Edit View Run Settings Add-ons Help



Run All

Markdown

Draft Session off (run a cell to start)



Task

You will use an LLM API to predict a score for each student answer.

The test data contains:

- `id`: the answer ID
- `question`: the assignment question
- `student_answer`: the answer that must be graded

Your submission file must contain exactly two columns:

- `id`
- `predicted_score`

Each predicted score must be between 0 and 5.

Main Part to Edit

You mainly need to edit the `PROMPT_TEMPLATE` variable in the section:

PROMPT ENGINEERING

Do not change the `id` values or the number of rows in the submission file.

+ Code

+ Markdown

點擊這裡開始執行程式

1. Get an API Key

獲得預測結果

```
# Wait before sending the next API request.
time.sleep(REQUEST_DELAY)

prediction_df = pd.read_csv(CHECKPOINT_PATH)
prediction_df.head()
```

/tmp/ipykernel_58/3766231481.py:34: FutureWarning: The behavior of DataFrame concatenation with empty or all-NA entries is deprecated. In a future version, this will no longer exclude empty or all-NA columns when determining the result dtypes. To retain the old behavior, exclude the relevant entries before the concat operation.

```
current_predictions = pd.concat(
[1/20] q5_001 -> 5.0
[2/20] q5_002 -> 5.0
[3/20] q5_003 -> 3.0
[4/20] q5_004 -> 5.0
[5/20] q5_005 -> 5.0
[6/20] q5_006 -> 5.0
[7/20] q5_007 -> 4.0
```

+ Code + Markdown

9. Create submission.csv

The submission file must have exactly these two columns:

```
id,predicted_score
```

```
submission = test_df[["id"]].merge(
    prediction_df[["id", "predicted_score"]],
    on="id",
    how="left",
    validate="one_to_one",
```

在下面可以看到預測
結果一個個出現

全部跑完了以後
會出現一個叫
submission.csv的文
檔，把它下載下
來

Notebook

Input

+ Add Input

Upload

COMPETITIONS

[T1] NTHU Summer Camp 0723

Output (92KiB / 19.5GiB)

/kaggle/working

.virtual_documents

prediction_checkpoint.csv

submission.csv

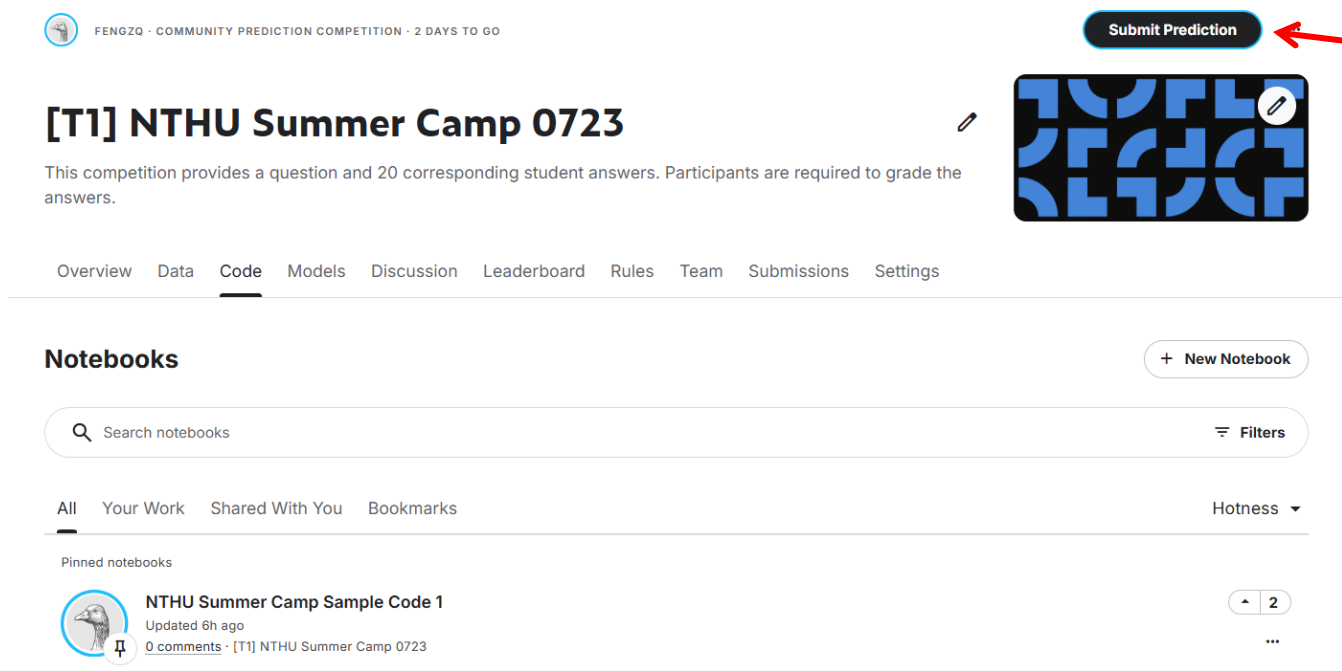
Table of contents

Task

Main Part to Edit

1. Get an API Key
2. Install the Required Library
3. Load the Test Set
4. Choose an API Provider
5. Prompt Engineering

上傳預測結果



FENGZQ · COMMUNITY PREDICTION COMPETITION · 2 DAYS TO GO

[T1] NTHU Summer Camp 0723

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Overview Data **Code** Models Discussion Leaderboard Rules Team Submissions Settings

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+ New Notebook

Search notebooks Filters

All Your Work Shared With You Bookmarks Hotness

Pinned notebooks

 **NTHU Summer Camp Sample Code 1**
Updated 6h ago
0 comments · [T1] NTHU Summer Camp 0723

把下載下來的submission.csv上傳到這裡即可

小練習第二題

PROMPT ENGINEERING 2

問題介紹

- 依舊是用LSTM (RNN) 模型做加減乘除題
- 題目: If we construct an training set using three-digit numbers while the evaluation set is constructed from two-digit numbers, what will happen to the quality of your answer generation?

翻譯: 如果用三位數運算訓練模型, 但是測試用兩位數運算來測試, 會發生什麼? 這題滿分10分

我們主要會看:

- 有沒有實驗結果
- 有沒有針對實驗結果的分析

給個例子

問：如果用三位數運算訓練模型，但是測試用兩位數運算來測試，會發生什麼？

答：讓模型透過訓練集學習較複雜的算法，而驗證集算提醒模型說你不能只會複雜的算法，你也要會能夠簡單算出來的方法，如果國小生問一個他目前學業上遇到的一個問題，而你卻用一個大學生解法教他的話他肯定會不理解。用這種方法提升模型泛化能力，模型看到只有2位數的輸入時還能知道說這是甚麼，而不是愣在一旁。

給個例子

問：如果用三位數運算訓練模型，但是測試用兩位數運算來測試，會發生什麼？

答：讓模型透過訓練集學習較複雜的算法，而驗證集算提醒模型說你不能只會複雜的算法，你也要會能夠簡單算出來的方法，如果國小生問一個他目前學業上遇到的一個問題，而你卻用一個大學生解法教他的話他肯定會不理解。用這種方法提升模型泛化能力，模型看到只有2位數的輸入時還能知道說這是甚麼，而不是愣在一旁。

這題給2分，原因是沒跑數據

再舉個例子

- 正確率幾乎歸零，降至0.08%以下，並且隨著訓練批次的增加，正確率並沒有顯著提升，這是由於LSTM本質上是模式辨識、訓練字元間相關性的模型，因此當我們的訓練資料皆是三位數，他就會預期該三位數後會接著一個運算符號或是括號，而換成兩位數後，這種模式就會徹底錯誤，導致儘管該模型也許能通過字元間相關性了解運算符的作用，也會由於該模式的破壞而出現錯誤導致正確率近乎歸零。

再舉個例子

- 正確率幾乎歸零，降至0.08%以下，並且隨著訓練批次的增加，正確率並沒有顯著提升，這是由於LSTM本質上是模式辨識、訓練字元間相關性的模型，因此當我們的訓練資料皆是三位數，他就會預期該三位數後會接著一個運算符號或是括號，而換成兩位數後，這種模式就會徹底錯誤，導致儘管該模型也許能通過字元間相關性了解運算符的作用，也會由於該模式的破壞而出現錯誤導致正確率近乎歸零。
- 這個答案10分

多輪PROMPT

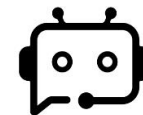
- 第二個小練習是第一個練習的升級版，要求各位設計兩輪的prompt

先舉個例子：



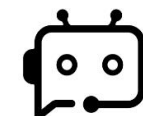
幫我打個分吧，題目是：XXXXXX

好，這個分數是X



你再好好想想是不是打分太高了？

喔確實，正確分數應該是X



一些可以參考的兩輪PROMPT思路

1. 讓模型反思自己的打分
 2. 把問題拆成兩個兩個步驟，先問一個，等模型回復了第一步以後再問下一個
 - 比如先讓模型判斷是否有回答一些你認為比較重要的部分，然後再打分
 3. 先讓模型輸出你給的範例哪個和你的input最像然後再打分
- ...

請各位設計兩步的PROMPT

NTHU Summer Camp Sample Code 2 4f0f3b Draft saved

File Edit View Run Settings Add-ons Help

+ - ✂ 📄 📁 ▶ ⏮ Run All Markdown ▾

● Draft Session (1m) H DD C V C R A M 🔌 ↺ ⋮

```
[9]: # =====  
# PROMPT TEMPLATE FOR TURN 1  
# =====  
  
PROMPT_TEMPLATE_1 = """  
You are an expert teacher evaluating a student's answer.  
  
Task:  
Analyze the student's answer step-by-step based on the provided question.  
  
Instructions:  
1. List the correct points made in the answer.  
2. Identify missing key information, incorrect statements, or flaws in reasoning.  
3. Just provide a preliminary score from 0 to 10 with a brief rationale.  
  
QUESTION:  
{question}  
  
STUDENT ANSWER:  
{student_answer}  
"""
```

+ Code + Markdown

```
[10]: # =====  
# PROMPT TEMPLATE FOR TURN 2  
# =====  
  
PROMPT_TEMPLATE_2 = """  
You are reviewing your previous draft evaluation of a student's answer.
```

找到中間的這個部分，
紅色字體的引號內可以
給各位設計。